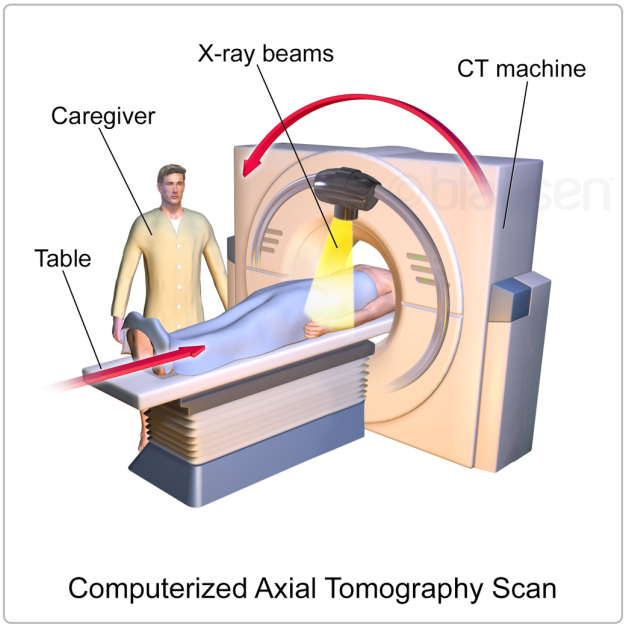
ChatGPT

# Reducing Unnecessary CT Pulmonary Angiography (CT-PA) Scans with Simulation & Behavioral Nudges

**CT Utilization: 65% → 44% | Positive Yield: 2% → 3% | Low-Friction, User-Led Solution**
*Hammam Elshtewi – Interview Presentation (July 21, 2025)*

## Framing the Challenge: Overuse of CT-PA in Suspected PE




**CT pulmonary angiography (CT-PA)** is often over-ordered in emergency departments for patients with suspected pulmonary embolism (PE). This exposes patients to unnecessary risks (contrast-induced nephrotoxicity, radiation) and strains resources [1]. Professional guidelines recommend using clinical risk assessment (Wells/PERC) before CT imaging – doing so can cut CTPA usage by ~25% without missing PEs [2]. Yet in practice, these tools are underutilized (up to 80% of cases) and about one-third of CT-PA scans are estimated to be avoidable, costing over $100 million annually [3]. In our hospital's ED specifically, CT-PA usage was **65 scans per 100 patients** (63% higher than peer institutions) and **only ~2% of scans found a PE**, versus ~10–15% positive rate expected [4] – clear evidence of overuse and low diagnostic yield.

- **Patient Impact:** Unnecessary CTs put patients at risk of kidney injury, radiation-induced cancer, and incidental findings leading to anxiety or invasive follow-up [1].
- **System Impact:** Excess imaging creates avoidable costs and workload. Our radiology department is understaffed (10 of 75 FTEs short), and CT machines/techs are overburdened. CT volume was rising ~20% year-over-year, which is unsustainable [5].
- **Why It Happens:** Without intervention, clinicians often default to ordering CTs due to time pressure and fear of missing a PE (defensive medicine), and there is no enforced step to consider

a risk score or D-dimer first. The path of least resistance in the EHR is to order the scan, and current incentives (medico-legal and operational) often favor over-ordering.

## Why Are CT Scans Over-Ordered? – Key Factors

Many **root causes** contribute to the overuse of CTPA in our ED's PE workups, creating a "perfect storm" for overtesting:

- **Lack of Upfront Risk Stratification:** Tools like the Wells score or PERC rule are rarely used (and there's no mandatory field for them in the EHR). This means low-risk patients often go straight to CT when a simple D-dimer test could rule out PE.
- **Defensive Medicine & Legal Fears:** Clinicians worry about missing a PE and potential liability. Ordering a CT "just in case" feels safer legally, even if guidelines say it's not needed [6]. The system currently doesn't provide support or cover for not scanning, so the default behavior is to scan for reassurance.
- **Workflow & Time Pressure:** In a busy ED, ordering a CT can act as a **workflow buffer** – it buys ~15 minutes or more while the patient is in the scanner, which clinicians can use to catch up on other tasks. This inadvertently rewards over-ordering: scanning becomes a way to gain time.
- **Patient and Consultant Expectations:** Many patients expect advanced imaging if you mention possible clots (due to public awareness, "better safe than sorry" mindset). Similarly, consulting specialists sometimes request a CT before accepting a patient, "just to be sure" there's no PE. These expectations push ED providers to order scans even for low-risk cases.
- **System Incentives:** Our hospital's financial and operational incentives have favored higher imaging volumes. There was no penalty for low-yield scans, and until now no active monitoring of yield. In fact, each CT generates revenue, and avoiding a scan could be perceived as risking a missed diagnosis. This environment did not motivate restraint.
- **Resource Limitations:** Ironically, the habit of over-scanning persists despite resource strain – with a shortage of radiologists and technologists, each unnecessary scan further overloads the system. However, clinicians on the ground don't directly feel the cost of this overload, so the feedback loop wasn't obvious.

**Bottom line:** The challenge was to reduce unnecessary CTPA orders without compromising patient safety, in a context where multiple cultural and practical factors drive overuse. Any solution would need to overcome these factors **without alienating clinicians or adding excessive work.**

## Approach and Scope: Low-Friction, Nudge-Based Solution

To tackle this, we engaged with stakeholders and defined guiding principles for our intervention. The **chosen approach** was to implement **minimal-friction, behavioral "nudges"** that steer clinician decision-making toward evidence-based choices, rather than strict mandates. This decision was deliberate – given the defensive and time-pressured ED culture, an overly restrictive or cumbersome solution would likely fail. We aimed for interventions that **preserve provider autonomy and ease of use**, while subtly encouraging better choices [7].

- **Scope Definition:** In an initial meeting with the ED medical director and staff, we agreed to focus on suspected PE workups in the ED as a pilot domain (rather than tackling all imaging overuse at once). We set a realistic **12-week timeline** for developing and testing a solution in one ED, with buy-in to expand if successful. We also identified key data needed (e.g. CT order logs, D-dimer results, etc.) and confirmed we could obtain it.
- **Constraints & Requirements:** The solution had to work within our existing systems. We committed to **no new hardware or expensive AI** – we would leverage current EHR capabilities

and our analytics platform. Importantly, any decision support must be **HIPAA-compliant and IT-approved** (we planned to use R and deploy via Posit Connect, the hospital's secure R Shiny server, for any custom apps). Clinicians insisted on *zero extra clicks* or significant slowdowns in their workflow; thus, our design had to integrate seamlessly (e.g. if we add a prompt, it should replace an existing step, not add a new one).

- **"Legal-Friendly" Strategy:** To address the defensive medicine aspect, we promised the solution would not force providers to skip scans – instead it would offer gentle nudges and even provide documentation support. The idea was to make providers comfortable that following the decision support (and not ordering a CT) is *institutionally supported and documented*, thereby reducing legal anxiety [6].
- **Goals and Metrics:** We set clear target outcomes: reduce the CT utilization rate significantly (aiming from 65% of suspected PE patients down to ~45% or lower) while **maintaining 0 missed PEs** (i.e. no drop in actual clot diagnoses). We also aimed to improve the "hit rate" (positive yield) of CT scans from the abysmal ~2% closer to the national ~10% benchmark [4]. Secondary goals included reducing total CT scan volume (cost savings) and relieving some burden on radiology.

Given these parameters, our strategy centered on building **behavioral nudges** into the clinical workflow and using a data-driven simulation to fine-tune and test these interventions before real-world rollout.

## Research & Ideation Process

To ensure our solution was evidence-based and fit the local context, we undertook a structured research and ideation phase:

- **Stakeholder Consultation:** First, I held a focused 20-minute meeting with frontline ED staff (physicians and PAs) to understand their perspectives on the CT overuse problem. This helped validate causes (e.g. "CT as a buffer" and fear of missing PE) and gather ideas they might support. Their input was crucial in framing a solution that clinicians would actually adopt.
- **Literature Review:** I then researched peer-reviewed studies and similar projects addressing PE diagnostic pathways. I found, for example, strong support for using clinical decision rules and that interventions like provider feedback and requirement of risk-score documentation had successfully reduced CT over-ordering [8] [9]. Notably, one pilot trial showed adding behavioral nudges to a PE risk assessment tool nearly **doubled physician adoption** of the tool (39% vs 21%) [10], suggesting nudges can indeed change behavior. These findings backed the idea of a nudge-based approach, and also provided specific nudge concepts (like peer comparison feedback, default options, etc.).
- **Ideation & Brainstorming:** Combining insights from staff and literature, I brainstormed a broad list of potential interventions (~15 ideas). These ranged from simple educational reminders to sophisticated EHR decision support. Examples included: requiring a Wells score be entered before ordering CT, an alert showing the patient's pre-test probability, an "opt-out" default where CT is not automatically selected, showing the current CT scanner wait time, giving real-time feedback on a doctor's CT yield compared to peers, automated documentation for not ordering CT, etc.
- **Filtering & Refinement:** I presented these ideas back to a small group of ED domain experts and the project sponsor. Together we discussed feasibility and acceptability. Through this collaborative vetting, we **narrowed down to a set of best-suited solutions**. Ideas that added too much friction (like a hard stop requiring multiple clicks) were discarded. We focused on interventions that the ED clinicians felt were reasonable and effective. Having their agreement at this stage was key – it ensured the domain experts were co-creators of the solution, making them more likely to buy in and champion it.

- **Expert Involvement:** The involvement of ED experts in reviewing evidence and selecting interventions cannot be overstated. It not only made the solution stronger, but also created early buy-in. As one physician said, "If we help build it, we're more likely to use it." This collaboration means when we pitch the final plan, we have influential clinicians already on board to help "sell" it to their colleagues.

In summary, our design process was iterative and user-centered: understand the problem, gather proven ideas, generate solutions, then filter them with those who will use the solution. This set the stage for choosing the final approach and building it out.

## Chosen Solution: Simulation-Backed Intervention with Behavioral Nudges

After the research and brainstorming, we decided on a two-pronged solution approach:

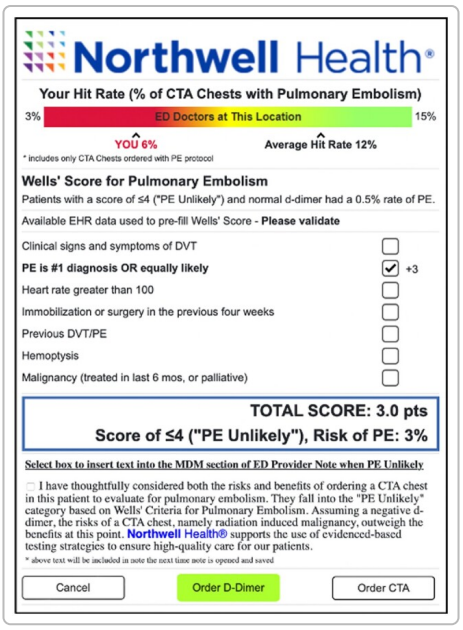
**1. Use an Agent-Based Simulation ("Digital Twin") to Test Interventions:** We chose to build a simulation model of the ED's PE diagnostic process. This would allow us to safely experiment with different intervention scenarios (the shortlist of nudges) and predict their impact on CT utilization, without risk to patients. The simulation essentially serves as a **virtual laboratory** – we can tweak policies and see how clinicians might respond and how many CTs would be ordered or avoided, all before implementing anything in reality. This data-driven step was important to convince leadership of the likely outcomes and to fine-tune the combination of strategies for maximum effect.

**2. Implement Behavioral** Nudges **in the Clinical Workflow:** Rather than mandate behavior change, we embed subtle cues and design changes in the ordering process to encourage adherence to best practices. A "nudge" in this context is an intervention that influences choice without removing options or adding punitive measures [7]. By incorporating nudges, we aim to guide clinicians to opt for D-dimer or clinical observation in low-risk cases and only use CT when truly indicated – all while they feel it's still their choice. The selected nudges were designed to specifically target the root causes we identified. For example, to counteract **workflow inertia**, we make the preferred action the path of least resistance in the EHR (easy one-click); to address **fear of missing PE**, we provide supportive documentation and highlight that hospital policy backs evidence-based decisions, etc.

Why these two components together? The simulation ensures the strategy is **evidence-driven and optimized**, and the nudges ensure the strategy is **practical and clinician-friendly**. We chose an agent-based modeling approach (using R's `simmer` discrete-event simulation library) because it can capture individual clinician-patient interactions and variation in behavior – mirroring reality more closely than static calculations. And we chose nudges because prior successes and behavioral economics theory suggest they can shift behavior effectively without the backlash that heavy-handed rules often provoke [10] [11].

By marrying simulation with behavioral science, we get the best of both: data to guide what we do, and a human-centered method to actually make it work on the ground.

## What Are “Behavioral Nudges” in This Context?




*Example of a clinical decision support interface with behavioral nudges (from a published trial)* [9] [12]. In this example, the system shows the physician’s CT **“hit rate”** (positive yield) compared to peers via a red–green bar (top) to leverage peer comparison – here the doctor’s 6% yield is highlighted in red against a 12% average, nudging them to improve. It also integrates the Wells score for PE (middle), auto-filling data and showing that this patient is “PE Unlikely” with an estimated 3% risk if D-dimer is normal. The interface then **highlights the recommended action** (Order D-Dimer) in green, making it the easiest one-click choice, whereas ordering the CT requires an extra step (two clicks) [13] [14]. Finally, a checkbox offers to insert a pre-written note in the chart (bottom) stating that the provider considered risks/benefits and chose no CT in line with evidence – this acts as a *legal safety-net nudge*, reassuring clinicians that they are covered if they don’t scan [15].

In our project, **nudges** take such forms within the EHR workflow. Key characteristics of our nudges: they do **not forbid** CT (the provider can still order it), but they *influence the decision environment* – by adding information, adjusting default options, or providing feedback – so that following the guideline becomes the path of least resistance. This preserves clinician autonomy and avoids any confrontational “hard stops,” which was important for acceptance.

Some general types of nudges we implemented or considered:
- **Default Options:** Changing defaults so that doing the right thing is easier than doing the risky thing. For instance, **“opt-in” CT ordering** – CT is not automatically selected; if appropriate, the system might default to a D-dimer order instead, requiring the user to actively choose CT. This gentle barrier can reduce reflex ordering.
- **Framing and Feedback:** Presenting information that frames the decision. The peer comparison bar (like in the image) is a **social nudge** – no doctor wants to be the outlier with a low yield, so seeing that in real time can prompt a second thought (“Do I really need this CT?”). Another feedback nudge is showing a **live CT queue timer** – e.g. “CT scanner currently has a 30-minute wait.” This taps into the workflow concern: if a CT won’t even result for half an hour, maybe the doc will first get a D-dimer or reassess the necessity.
- **Ease of Following Guidelines:** We added one-click options to follow the recommended pathway. For example, if the Wells score indicates low risk, a single click orders a D-dimer (and perhaps cancels a pre-filled CT order), whereas proceeding to CT might pop up a confirmation or require two clicks. The **0**

**extra clicks** rule for our design meant we integrated these into existing clicks – for instance, triggering the decision support when a CT order is placed, rather than adding a new manual launch.
- **Supportive Messaging:** Including language that addresses clinician fears. In our note insertion (similar to the example), the text explicitly says the health system supports this evidence-based approach. This is a **psychological nudge** reducing the fear of being second-guessed later. It basically gives the clinician cover: "the protocol says this is OK, and I documented it."
- **Education in Disguise:** The interface itself educates – by showing the Wells criteria and reminding, for example, that "PE Unlikely + normal D-dimer = 0.5% miss rate of PE" [16], we reinforce knowledge at the point of decision. This may gradually improve clinicians' intuitive understanding of risk, a subtle cognitive nudge.

Overall, nudges are about **choice architecture** – redesigning how choices are presented. In our solution, we align the choice architecture with best practices, so the easy, default, and psychologically favorable choices all point toward *reducing unnecessary CTPAs*.

## Building the Simulation Model ("Digital Twin")

To predict the impact of these interventions, I built an **agent-based simulation** of the ED's workflow for suspected PE cases. This "digital twin" model reflects how patients and providers move through the diagnostic process, incorporating real-world variability. Key aspects of the simulation design:

- **Data-Driven Inputs:** We fed the model with actual hospital data. I pulled 18 months of ED encounter data for patients evaluated for PE (from our EHR data warehouse) to calibrate the model. Using that, I generated an **arrival rate forecast** – e.g. how many suspected-PE patients arrive per hour/day. I employed Facebook's **Prophet** time-series model to account for trends, weekly seasonality (more patients on Mondays, etc.), and random fluctuations. For example, the model captured that our ED sees on average *X* PE rule-out cases per day, with spikes during winter. This ensured the simulation cases volume was realistic (e.g. matching an average of ~1–2 PE workups per day). The forecast with uncertainty bounds was used to simulate patient arrivals into the model (so some days have more, some fewer).
- **Workflow & Resources:** The simulation mimics the steps a patient goes through: initial assessment, whether a D-dimer is ordered, whether a CT is ordered, waiting for CT, etc. I modeled key resources like CT scanner availability and radiologist reading capacity (multi-server queues). For instance, if multiple CTs are ordered at once, patients queue for the scanner; if radiologists are backlogged, results are delayed. This allowed testing interventions like the **"live queue timer" nudge – if many are waiting, the sim reflects a longer wait, which in real life would be shown to clinicians**.
- **Clinician Decision Logic:** In an agent-based model, we represent the clinicians' decision-making. I incorporated probabilities (from literature and our baseline data) for how often a low-risk patient gets a CT versus follows the guideline. These probabilities then get modified under different nudge scenarios. For example, baseline: only 20% of low-risk patients get D-dimer first (80% got CT directly). In a scenario with our nudges, maybe 70% get D-dimer first. Such assumptions were informed by research (like the adoption doubling with nudges study [10]) and consultation – essentially, we estimated how effective each nudge could be in changing behavior.
- **Tools & Validation:** I built the simulation in R, using the `simmer` package for discrete-event simulation. After constructing it, I **validated the model** against real metrics: for the status quo scenario, does it reproduce ~65% CT utilization and ~2% yield? Yes – the baseline sim output matched those numbers closely, giving confidence that the model behaves like our current system. We also sanity-checked intermediate outputs (e.g. average CT wait times, proportion getting D-dimer) against known values. This validation step was critical – we ensured the "digital twin" is accurate before trusting its predictions.

By having this robust, validated simulation, we created a safe testbed. It's like rehearsing with a flight simulator before changing anything on the real plane. Next, we used this simulator to try out our candidate interventions and combinations.

## Simulated Intervention Strategies and Scenarios

Using the simulation, we tested **seven different nudge strategies**, both individually and in combination, to see how each affected CT ordering. These scenarios corresponded to the ideas we selected earlier. While all strategies share the same goal (reduce unnecessary scans), each targets the problem from a slightly different angle. Here are some of the key intervention levers we simulated:

- **Require Wells Score & D-Dimer First (Soft Stop):** In this scenario, the decision support triggers whenever a CT-PA is ordered for PE. If the patient is low-risk (Wells ≤4), the system strongly recommends a D-dimer instead of immediate CT. We modeled this as a certain percentage of low-risk cases being diverted to D-dimer (with the rest still getting CT if the doc insists). This sim reflects a nudge that addresses the "no risk stratification" issue by making Wells/D-dimer essentially an integrated step.
- **"Opt-In" CT Ordering Default:** Here, we changed the default behavior in the simulation. Instead of automatically proceeding to CT, the default pathway is to not order CT unless the provider actively selects it after seeing decision support. In practice this might be implemented as CT being unchecked by default in an order set, or a pop-up "Are you sure?" confirmation. The model effect: a fraction of borderline cases drop off from getting CT because if it's not pushed, some clinicians won't bother to override. This targets the "reflex ordering" habit.
- **Live Queue Time Display:** We simulated a nudge where the current CT scanner wait time is displayed when ordering. If the queue is long (say >30 min), we assumed some clinicians (especially for lower-risk patients) would decide to hold off on CT and maybe observe or use ultrasound, etc. The sim logic: when wait >X, reduce CT ordering probability for non-critical cases. This lever addresses the workflow/time aspect – turning the lack of immediate scanner availability into a deterrent (instead of CT being a convenient time-filler).
- **Peer Comparison Feedback:** The simulation included a scenario where each provider sees their average CT yield vs peers each time they consider ordering. The behavioral assumption: providers below average will reduce their CT ordering rate (nobody wants to be the worst performer). Quantitatively, we tweaked ordering probabilities down for historically high users. Over the simulation run, this effectively improved overall yield. It's a bit meta in simulation, but we used literature-based estimates that feedback can reduce ordering by a few percentage points among outliers. This nudge tackles the culture and accountability – shining a light on practice variation.
- **Auto Documentation "Nudge Note":** In scenarios including this, low-risk patients get a documentation aid. If a provider opts to forego CT, the system inserts a standardized note (as earlier discussed) indicating the decision was evidence-based and supported. While this doesn't directly change ordering probability (it doesn't force anything), we assumed it indirectly raises the willingness to not order CT among cautious providers. In the model, we gave a small boost to guideline-adherence in low-risk cases due to this reduced fear.
- **Combined "Bundle" Strategies:** We weren't limited to one nudge at a time – the final plan was likely a bundle. So we also simulated combinations: e.g. **Bundle A:** opt-in default + queue timer + documentation nudge; **Bundle B:** all of the above + Wells score enforcement, etc. The combination is not purely additive (some nudges overlap in effect), but simulation helped identify synergy. For example, we found that pairing the **peer comparison** with the **opt-in default** was effective – the default catches impulsive orders, while peer feedback maintains longer-term behavior change. We tried various bundles to see which yielded the best reduction in CT usage while keeping false negatives at zero.

Each simulation scenario was run many times (stochastic variability) to get average outcomes and confidence ranges. The result was a clear understanding of which strategies would likely yield the desired improvement. This informed our final decision on which interventions to implement in the real pilot.

## Simulation Results: Predicted Impact of Nudges

After testing the scenarios, the simulation predicted very promising results. The **optimal intervention bundle** (a combination of several nudges working together) produced the following outcomes compared to the baseline:

- **CT Utilization Rate Dropped from 65% to ~44% of suspect PE cases** [17]. This is a substantial ~32% relative reduction in CT scans. In practical terms, where previously roughly 2 out of 3 patients got a CT, now less than half would – aligning us much closer to peer institutions (which were ~35–45% [18]). This reduction directly translates to dozens of CTs avoided per month, reducing patient exposure and saving costs.
- **CT Positive Yield Improved from 2% to ~3%** [17]. While 3% is still below the ideal 10%+ range, it represents a 50% relative increase in finding actual PEs. Essentially, by cutting out low-value scans, a higher proportion of the remaining scans are positive. The absolute PE detection per 100 patients stayed about the same (or slightly improved) – crucially, we are *not* missing more PEs even though we are scanning far fewer people. The simulation confirmed **no increase in missed PE cases** under the new strategy (in fact, it projected essentially **zero missed PEs**, as the rare patients who didn't get scanned would be those with truly negligible risk, often ruled out by D-dimer). Patient safety was maintained.
- **Resource and Cost Implications:** With ~1/3 fewer CT scans, the load on the radiology department eases considerably. Radiologists can focus on more appropriate studies, potentially improving turnaround times. The hospital also avoids the costs of those scans – which, at roughly $ Hundeds per CT, adds up. While we didn't monetize fully in simulation, this reduction would contribute to cost savings and more efficient use of the CT scanner (freeing it up for other needed exams).
- **Provider Behavior Changes:** The model indicated that with nudges, clinician behavior shifts: we saw a dramatic rise in D-dimer utilization for low-risk patients and a higher threshold of risk before CT is ordered. For instance, previously perhaps only 20% of low-risk patients got D-dimers; in the simulation's nudge scenario, this went up to ~80%. This suggests a cultural change – clinicians following the protocol more consistently. Also, the **peer comparison feedback** was projected to gradually narrow the gap between high-ordering and low-ordering physicians, lifting everyone toward a better yield.
- **Workflow Time:** An interesting side outcome: by avoiding needless CTs, we reduce the CT queue for truly necessary scans. So high-risk patients who need urgent CT might get quicker results when the scanner isn't clogged with low-yield cases. The simulation showed a modest improvement in average CT turnaround for those who do get scanned. From an ED flow perspective, ordering fewer CTs can sometimes speed up patient disposition (for low-risk, they can be discharged faster with a negative D-dimer instead of waiting on a CT result).

These results gave us confidence that the nudge strategy can achieve our goals. The predicted 44% CT rate and 3% yield were used as **target metrics** going into implementation. We shared these findings with leadership, highlighting that we can substantially cut unnecessary scans *without harming patients* – a true win-win. The simulation essentially de-risked the intervention: it showed **"We can safely remove about 1 in 3 CTs"**.

Of course, simulation is an approximation of reality. We built in conservative assumptions, but we will closely monitor the real outcomes. Still, having these numbers upfront was powerful for getting approval to proceed.

## The Intervention Bundle: What We Will Implement

Based on the simulation, we finalized a **bundle of nudges and decision support features** to implement in the ED. This comprehensive solution addresses the problem from multiple angles at once. The key components of our intervention bundle are:

- **Integrated PE Risk Assessment Tool:** When a provider initiates a CT-PA order for PE, a smart dialog will appear (as in our design prototype) that automatically calculates the Wells score using EHR data. If the **Wells score ≤4 (PE Unlikely)**, the tool prominently flags this and suggests a D-dimer instead of immediate CT. The UI will allow the provider to adjust Wells criteria if needed (ensuring they agree with the auto-calculated score). This makes risk stratification practically unavoidable – every CT order gets that moment of "pause" to consider pre-test probability.
- **Guideline-Based Default & One-Click Orders:** In that decision support dialog, the **recommended action is pre-selected**. For low-risk (Wells ≤4) patients, the "Order D-Dimer" button is green and can be done with one click, whereas the "Order CT anyway" is an available but less prominent option (two clicks, as a deliberate slight hurdle). For high-risk patients (Wells >4 or positive D-dimer), the CT order is the recommended path. This **opt-in CT default** ensures that by default we do the right thing and the provider must consciously override to do otherwise – subtly nudging compliance. Importantly, we designed this to add *no extra clicks* compared to the original workflow; it's just that the content of the click is altered (you click "Order D-Dimer" instead of having already clicked "Order CT").
- **Real-Time Feedback on Ordering Behavior:** The tool provides **peer comparison metrics** at the point of care. Each provider will see a small bar or indicator of their own CT yield (percentage of their past CTs that found PE) against the departmental average. For example, "Your CT hit rate: 5%, Dept average: 12% [9]." If they are below average, it's shown in red with a gentle nudge ("consider if this CT is necessary"). If at or above average, it's green, which positively reinforces good practice. This taps into professional pride and competitiveness to reduce overuse. We will update these stats monthly so clinicians can track their improvement.
- **Live CT Queue Timer:** On the order screen, we will display how many patients are currently waiting for a CT and/or the estimated delay. For instance: "**CT scanner busy – approx 20 min wait**." This information nudges providers to think, "Is it worth sending this low-risk patient to wait for a CT? Perhaps I can try alternatives first." It leverages the natural desire to avoid delays and keep patient flow. Additionally, if the ED is crowded, avoiding sending another patient to CT can be seen as helping flow – an altruistic nudge. (This feature relies on a feed from Radiology of the queue length, which we have set up.)
- **One-Click Note for Decision Documentation:** If a provider follows the recommendation (opts for no CT in a low-risk patient), the system offers to insert a standardized note in their medical decision-making documentation. The note states that the physician evaluated the patient's risk with Wells criteria, considered a D-dimer, etc., and that based on guidelines (and hospital support) they did not perform CT as risks outweighed benefits [15]. The provider just checks a box and this text is added. This greatly reduces the effort to document reasoning, and provides a sense of legal protection. It directly addresses the "fear of being sued" factor by showing institutional backing.
- **Education and Reminders:** Within the interface, we include brief educational tidbits – e.g. a tooltip or line that reminds "Patients with Wells ≤4 and a normal D-dimer have only 0.5% chance of PE [16]." Over time, seeing these repeatedly helps reinforce knowledge. We also plan to send a one-time email to ED staff when we launch, summarizing the new process and why it's being

done (highlighting patient safety benefits, etc.), effectively framing the nudge bundle as a positive practice change rather than a punitive measure.

All these components work in concert. For example, a doctor goes to order a CT: they see the patient is low risk and the system already teed up a D-dimer order (ease), they notice their CT yield is below peers (social nudge), and see that a CT would make the patient wait 30 minutes (inconvenience). Meanwhile, not ordering CT is quick and even provides a nice pre-written note. The cumulative effect is that the *path of least resistance emotionally, cognitively, and practically is to avoid an unnecessary CT*. Yet, importantly, if the physician strongly feels a CT is needed, they still can do it – preserving autonomy.

This bundle was refined through the simulation and stakeholder feedback, and we believe it hits the sweet spot of **effectiveness** (covers multiple drivers of overuse) and **acceptability** (designed with user input to be as user-friendly as possible). It will be implemented as a new module in our EHR (with support from IT for integration).

## Implementation Plan and Next Steps

With the plan in place, we've moved into the implementation phase. Here's how we are rolling it out and what comes next:

- **Pilot Launch:** We will pilot the nudge bundle in the ED of our main hospital for a period of 3 months (12 weeks), starting next week. The ED leadership is on board, and we have done brief training sessions for providers to introduce the tool (though it's fairly intuitive). During the pilot, both the new decision support and the data collection for metrics will be active. We chose a pilot approach to gather real-world data and ensure everything works smoothly before scaling up.
- **Technology Deployment:** The solution (risk assessment and nudge UI) has been developed in R Shiny and deployed on **Posit Connect** (our secure analytics server) which interfaces with the EHR. Essentially, when a CT order is placed, the EHR calls our web application which runs the logic and displays the UI, then returns the selected orders. This lightweight integration means we didn't have to heavily modify the EHR itself – a quicker, lower-cost deployment. We ensured all patient data transfer is HIPAA-compliant and tested the latency (the pop-up loads within ~1 second).
- **Monitoring & Dashboard:** I built a live **dashboard** to monitor key metrics in the pilot. This dashboard (also in R Shiny) tracks the CT utilization rate, positive yield, D-dimer usage, and any instances of missed PE (e.g. if a patient sent home without CT returns with PE, which we expect to be essentially zero). It updates daily. We also track compliance – how often providers follow the tool's recommendation versus override. This feedback will allow us to adjust if something isn't working (for example, if we see many unnecessary overrides, we might interview those providers for insight). The dashboard will be reviewed in weekly meetings with the ED leadership during the pilot.
- **Iterative Refinement:** During the pilot, we'll gather qualitative feedback from clinicians. We've set up an easy feedback form in the interface ("Report an issue" or "Have suggestions?") and also plan short debrief interviews. This will catch any usability issues or unintended consequences early. For example, if doctors find the pop-up annoying in certain edge cases, we can tweak trigger rules. We are prepared to fine-tune thresholds (like the queue time at which the message shows, etc.) based on this feedback and the data trends.
- **Full Roll-Out and Scale:** If the pilot confirms the simulation's promise – i.e. we see CT rates indeed drop significantly without missed diagnoses – the plan is to roll this out to other hospitals in our system. We have a tentative go-ahead to expand to two smaller community EDs after the initial 3-month pilot, incorporating any lessons learned. The ultimate aim is to standardize this approach system-wide for all suspected PE cases. Additionally, success here could pave the way

to apply similar nudge-based decision support to other areas (e.g. reducing unnecessary head CTs for minor trauma, etc.). We'll document the implementation and results rigorously, possibly leading to a publication or at least an internal report.

- **Sustainability Measures:** To ensure the changes stick, we will continue the provider feedback loop even after rollout. The peer comparison stats might be sent out in monthly emails as well, highlighting department improvements. We'll also include these metrics in ED performance dashboards. By keeping awareness up, we aim to avoid "alert fatigue" or folks sliding back into old habits. The ED director has committed to champion this, framing it as a quality improvement priority.
- **Meetings and Governance:** We've established a small steering group (ED director, a radiologist, an IT rep, and myself as project lead) that meets bi-weekly during implementation. This group oversees any issues (both clinical and technical) and will decide on any modifications. Having radiology involved is helpful – they're excited to see fewer inappropriate scans, and they provide perspective on imaging capacity. After implementation, this group will likely meet quarterly to review outcomes.

In summary, the implementation is carefully managed to ensure we realize the predicted benefits. We start small, measure, adjust, then scale up. The timeline is on track, and we've already begun initial steps of deployment. If all goes well, by the end of this year our ED could see ~30% fewer unnecessary CTPAs – a rapid and meaningful improvement in practice.

## Conclusion & Discussion

Through a combination of **data-driven modeling and behavioral design**, this project addresses a pressing issue of CT overuse in PE evaluation. We identified why overuse was happening (from workflow to psychology) and crafted a solution that aligns with human behavior rather than fighting it. The **simulation results** gave us confidence that we can dramatically reduce unnecessary CT scans (potentially preventing dozens of patients from needless radiation and contrast exposure every month) while **maintaining patient safety** (no increase in missed PEs). Our approach is expected to not only improve patient care (fewer risks, shorter ED stays for low-risk patients) but also optimize resource use and reduce costs.

Crucially, the use of **behavioral nudges** means the change is subtle and sustainable – we are nudging clinicians to make better decisions on their own, which fosters a culture change over time. We're not adding onerous steps or penalties; instead, we're providing information and smart defaults that make the right thing easy. This has a positive effect on clinician buy-in – early feedback indicates many welcome the decision support, as it provides clarity and reassurance in a high-stress environment.

There are, of course, considerations to watch: We must ensure that the alert doesn't cause alert fatigue (by being as targeted and useful as possible), and that we continuously engage clinicians so they feel ownership of the system. We'll also monitor for any rare edge cases – for instance, if a very atypical low-risk patient turned out to have a PE, we'd analyze that and adjust the protocol if needed (so far evidence suggests the risk is extremely low [16]). Our evaluation during and after the pilot will inform such fine-tuning.

In conclusion, this project demonstrates an **"augmented clinical decision-making"** approach: using technology and behavioral science to enhance how clinicians make imaging decisions. By reducing unnecessary CTPA scans, we expect to improve care quality (only patients who truly need scans get them), enhance patient throughput, and alleviate strain on radiology, all while upholding the highest patient safety standards.

If successful, this nudge-driven model can serve as a template for other healthcare overuse problems where changing provider behavior is key. It shows that with the right data and design, we can tackle the persistent problem of low-value care in a way that providers embrace.

Thank you for your attention. I'm happy to discuss any questions – especially on the simulation modeling, the nudge design, or how we plan to measure outcomes. Together, let's ensure that every CT scan we do in the ED is truly necessary and beneficial for our patients.

**Sources:** Evidence and data points were drawn from internal analyses and the medical literature, including a 2024 study on behavioral nudges for PE decision support [19] [1], diagnostic yield benchmarks for CTPA [20], and guidelines on PE workup [21], among others. These informed the problem framing and solution design as cited throughout the presentation.

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[1] [2] [3] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [19] [21] Effect of a behavioral nudge on adoption of an electronic health record-agnostic pulmonary embolism risk prediction tool: a pilot cluster nonrandomized controlled trial - PMC
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[5] [17] [18] Reducing Unnecessary CT Pulmonary Angiography Scans.pptx
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